Pappuri Prahladha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.88
Narayana Junior College, Kanuru	Class XII (Board of Intermediate Education, Andhra Pradesh)	2024	96.90%
Narayana EM high school, Badvel	Class X (Board of Secondary Education, Andhra Pradesh)	2022	93.67%

SKILLS

Languages - Python, C, C++, HTML/CSS, Bash, Shell Scripting

Frameworks & Libraries - PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, SciPy, OpenCV, Pillow, NLTK, FastAPI, Streamlit

Version Control - Git, GitHub

Operating Systems - Linux, Windows, macOS

Development Tools - Cursor, Jupyter Notebook, Google Colab, Claude Code

Data Annotation - CVAT

Core Concepts - Object-Oriented Programming (OOP)

PROJECTS

LLM From Scratch - Built a GPT-style Large Language Model from scratch by implementing a custom autograd engine, transformer architecture, self-attention, tokenization, embeddings, positional encoding, and autoregressive text generation.
Deep Learning Architectures From Scratch - Implemented CNN primitives, PyTorch CIFAR-10 training pipeline, RNN/LSTM sequence models, Transformer encoder with multi-head attention, and long-context modeling experiments.
From-Scratch ML Library (Fashion-MNIST) - Developed a reusable machine learning library featuring dataset utilities, preprocessing, Naive Bayes, LDA, Logistic Regression, Perceptron, Bayesian & Ridge Regression, and a minimal neural network framework with autograd.
Neural Network Foundations From Scratch - Built neural network implementations including XOR backpropagation, ADALINE, MLP architecture studies, optimization experiments, and regularization analysis with visualized results.
Explainable AI Fairwashing Detector - Developed a modular framework for analyzing fairwashing and explanation robustness using LIME, SHAP, surrogate models, and interpretation fragility detection techniques.
MLP Regression & CNN Feature Visualization - Implemented neural regression, convolution kernel visualization, feature-map analysis, pooling/stride experiments, and CNN training performance visualization using PyTorch.
Unsupervised Learning: PCA, GMM & Clustering - Compared PCA, K-Means, and Gaussian Mixture Models for dimensionality reduction and clustering through analytical experiments and visualizations.
Neural Network Regression & KNN Analysis - Compared neural network regression with K-Nearest Neighbors using error analysis, loss tracking, and visualization across different model configurations.
Bayesian Regression & Posterior Modeling - Implemented Bayesian regression with sequential posterior updates, MAP estimation, predictive uncertainty analysis, and probabilistic visualization.
Wordle Probability Strategy Simulator - Developed a probability-based Wordle strategy simulator using inverse-transform sampling, empirical CDF analysis, and letter-frequency statistics.
Content-Based Movie Recommender - Built a TF-IDF and cosine similarity based recommendation engine supporting automatic metadata processing and top-k movie recommendations.
Eigenvalue Solver in C - Implemented an eigenvalue solver in C with mathematical derivations documented using LaTeX and Markdown.
EPOCH Data Analytics - Performed exploratory data analysis using Pandas, NumPy, and Matplotlib, including preprocessing, statistical analysis, and visualization.
Modern AI Foundations Knowledge Base - Created a structured collection of AI study notes covering machine learning, neural networks, search algorithms, decision making, CNNs, and game AI concepts.

RELEVANT COURSES

Mathematics - Discrete Mathematics, Linear Algebra, Calculus I & II, Vector Calculus, Probability & Random Variables, Optimization(I & II)
Artificial Intelligence - Foundations of Machine Learning, Deep Learning, Programming for AI, Introduction to Modern AI
Computer Science - Introduction to Programming, DSA, Algorithms, DBMS(I), OOP
Other - Modern Physics, Control Systems, Linear Systems & Signal Processing

EXTRACURRICULAR

- Regularly explore the latest AI research papers, open-source models, and developer tools; implement concepts through hands-on experiments and prototype development.
- Build practical software solutions for real-world productivity challenges using AI-assisted development workflows, emphasizing rapid prototyping and iterative refinement.
- Developed **WinLaunch**, a lightweight Windows application launcher optimized for fast keyboard-driven application switching and enhanced productivity.
- Developed **Screen Annotator**, a high-performance Windows screen annotation tool inspired by Presentify, featuring an improved user interface and optimized performance.
- Continuously evaluate emerging AI frameworks, large language models (LLMs), and automation tools by building experimental applications and proof-of-concept projects.
- Competitive Programming: **Codeforces** maximum rating **1421**.